AA 272 Fall 2025. Homework 3

YOUR NAME, YOUR_EMAIL@stanford.edu

Instructors: Prof. Grace Gao, Keidai Iiyama, Guillem Casadesus Vila, and Hannah Nabavi

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Goal of this problem set: Until this point, we assumed that we were given the satellite position, clock bias, and pseudorange. But how do we calculate these quantities? In this problem set, you will learn to calculate each part and assess the relative importance of different complications.

Instructions & Notes

1. This problem set is **due at 4:00 PM on October 31, 2025 via Gradescope**.
2. This homework is scored out of **50 points total** and includes a corresponding self-care component. The point value for each homework question is indicated next to the question number. Note that each homework will be equally weighted in the calculation of your final grade.
3. Please note that homework is intended to supplement and reinforce your learning. We design them to guide you through the process of going deeper into the concepts we cover in class, and beyond.
4. You are free to discuss class contents and these problems with others. However, your submitted solutions and code must be your own work, written from scratch.
5. **AI tools** (e.g., Chat GPT, Gemini) can be viewed as search engines or even classmates. They must not be used to directly solve any of the problems. For example, it is acceptable to discuss general topics about GNSS, edit a plot, or how so solve a linear system using `numpy`.
6. **Ensure that you typeset your solutions**. We encourage using the provided notebooks or LaTeX to type up your solutions, but you are free to use Microsoft Word or other editors.
7. Ensure that your solution includes the plots required by the question, **in the specified format** outlined in the problem for full credit.
8. If you are spending an excessive amount of time on any part of any question, please post on Ed or come to office hours (see Canvas).
9. Please **start early** so you can address any questions you might have, and have fun!

Please submit both a PDF of your solutions and your code via the programming assignment entry on Gradescope.

Problem 1: Orbits and Ephemeris [26 points]

Goal of this problem: Students will use GNSS measurements collected on an Android phone and broadcasted ephemeris data to compute the satellite positions and clock biases. Students will assess the relative importance of different corrections to the satellite orbit and clock.

Our pseudorange model is given in Equation 1 below.

$$\rho^{(k)} = \sqrt{(x^{(k)} - x)^2 + (y^{(k)} - y)^2 + (z^{(k)} - z)^2} + b_u - B^{(k)} + \epsilon^{(k)}$$

So far, we assumed that we were given the satellite position $(x^{(k)}, y^{(k)}, z^{(k)})$ and satellite clock bias $(B^{(k)})$. Behind the scenes in GNSS Lib Py, you accessed *precise* ephemeris data in problem 2 from homework 1. These precise ephemeris are *post-processed.* Similarly, in problem 1 from homework 2 and in our in-class activity in lecture 6, the data we used already included the GPS satellite positions and clock biases. But how do you get the satellite information in real time? For that, you need the *broadcast ephemeris.* In this problem, you will implement parts of the algorithm from lecture 8 to obtain these satellite positions from the ephemeris parameters. We also included the calculation of the satellite velocity in the code base, which you would need for Doppler calculations and more advanced localization approaches. To obtain the satellite clock biases, you will also get to follow and implement the algorithm outlined in the GPS Interface Specification itself!

Please upload these notebooks to Google Drive to use Google Colab or work on your computer if preferred, and post in Ed if you have any issues!

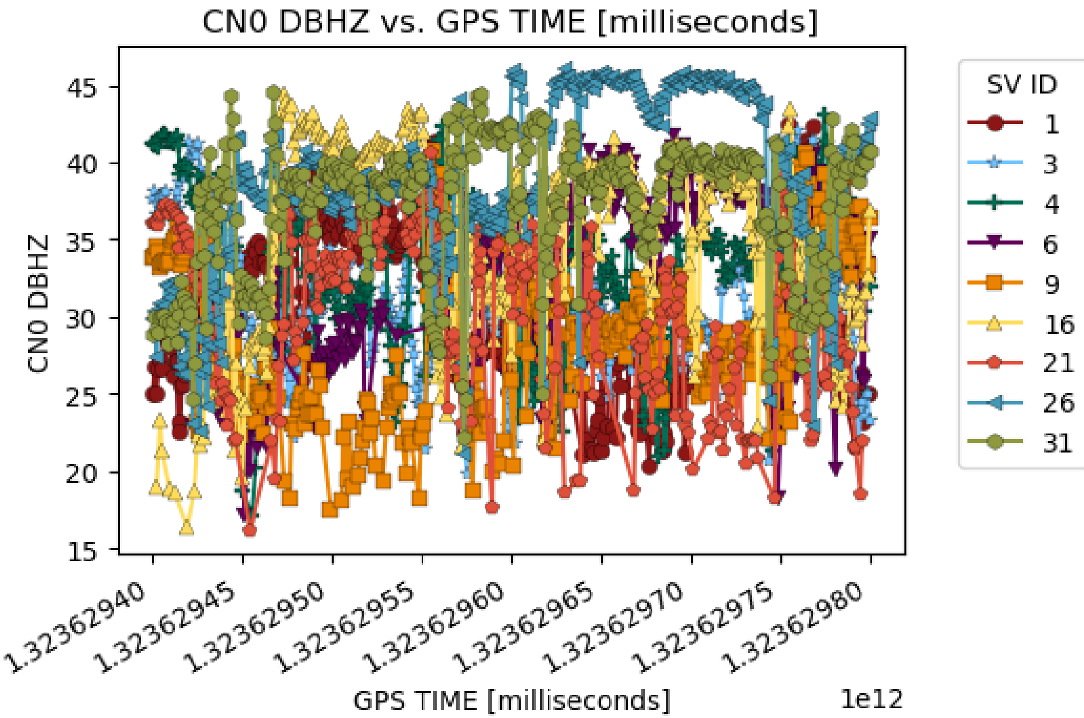
Setup

1. Set up workspace
2. Check that files are uploaded
3. Install dependencies
4. Load receiver data

Specifically, the data is from a Google Pixel 5 phone in the 2021-12-US-MTV-1 dataset from the 2022 Google Smartphone Decimeter Challenge. As you might guess from the name, the data is from a receiver driving near Mountain View, CA. Note that the file includes both raw and post-processed data. For this problem set, we are mainly interested in the timestamps of the receiver.

Let's convert the position obtained using WLS to latitude, longitude, and altitude to visualize the receiver's trajectory.

We can also visualize some of the data, such as the C/N_0. Note that `gnss-lib-py` provides several convenient plotting functions.



5. Download the satellite data In homework 1, you used the *precise* ephemeris, which are post-processed. However, in real-time, you only have access to the *broadcast* ephemeris made available in the satellite navigation message or retreivable from cellular towers via Assisted-GPS (A-GPS) services.

For this homework, we are accessing the broadcast ephemeris. This comes in a Receiver Independent Exchange (RINEX) format, maintained by the International GNSS Service (IGS) that you may recall from homework 1. Let's download the broadcast ephemeris and inspect it.

Now that we have both the receiver and satellite data, we are ready to start the problem!

1.1 Kepler's Equation with Newton-Raphson [4 points]

In lecture 5, we discussed that we can compute the eccentric anomaly E for the satellite orbit through Kepler's Equation if we have the orbit's mean anomaly M and its eccentricity e.

$$M = E - e \sin E$$

However, because Kepler's Equation has no closed-form solution, we need to solve for the eccentric anomaly E iteratively. How would you use Newton-Raphson to iteratively solve for E (given M and e)? Write your answer in a clear, step-by-step approach. Ensure you include a clear mathematical expression for the iterative update step, which corrects your previous guess of the eccentric anomaly. For your initial guess of the eccentric anomaly, E_0, choose M.

Hint: Besides lecture 5, the introduction of these course notes might be helpful to reference when structuring the Newton-Raphson problem and algorithm.

ANSWER

[Provide a mathematical derivation of Newton-Raphson for solving Kepler's Equation and implement the function below.]

We provide a function that computes the eccentric anomaly from ephemeris parameters, using the function that you just implemented.

1.2 Satellite Clock Bias [4 points]

We also have to compute the satellite's clock bias term, corresponing to $B^{(k)}$ in our pseudorange equation. Refer to the most recent GPS Interface Specification (IS) under GPS Space Segment/Navigation User Interfaces, and refer to the section on SV Clock Correction. You will notice the clock bias expression has four additive terms, one of which is a relativistic correction term. In addition to these terms, there is an additional fifth correction called the group delay term (t_{GD}, labeled in the code as `TGD`), that we already included for you.

Hints:

a. Your clock bias computation from following the GPS IS will give you clock corrections in seconds, but we already provide the code to convert to meters.

b. The GPS IS has a table that defines all clock and ephemeris parameters and their corresponding symbol representation used by the IS.

c. Notice that we want to use the time offset to account for the clock drift.

d. For the relativistic term, all the terms are multiplied together; there is no exponentiation. The formatting in the GPS IS is just a bit mis-formatted.

ANSWER

[Provide the formula for the satellite clock bias correction and implement the function below]

Below we provide a function that computes the satellite clock delay, using the function that you just implemented.

1.3 Ephemeris Errors [2 points]

After implementing a solver for Kepler's equation and a function to compute the clock corrections, we are ready to take the orbital parameters in the ephemeris data and compute the satellite states. The term *states* refers to the combination of position, velocity, and clock bias in the ECEF frame. The function below implements the algorithm introduced in lecture 5, which is common in GPS receivers. Note that it calls the functions that you just implemented.

Now we will loop through the receiver data and run the functions above. Note that the implemented functions allow accounting for different effects, such as whether to use the relativistic clock corrections. Below we introduce a configuration dictionary and helper functions so that you can more easily separate out each effect.

We use GNSS Lib Py to generate some of the plots you will need for the homework to avoid too much work on your end. The `T` and `F` characters at the start tell you which corrections were turned on or off.

Set all the toggles to `True`. This will match the algorithm in the GPS Interface Specification. We will compare the satellite positions, velocities, and clock bias that Google computed for the 2022 Google Smartphone Decimeter Challenge. Please include your plot for the error in the x ECEF position `x_err_m` and the clock bias `b_err_m`.

Side note: Google is likely doing a bit more processing than we are, so it will not line up exactly, but the errors should be close to zero.

ANSWER

[Complete the cell below]

```
[ ]: sv_state_errors.rows
```

```
[ ]: ['gps_millis',
      'sv_id',
      'gnss_id',
      'x_err_m',
      'y_err_m',
      'z_err_m',
      'vx_err_mps',
      'vy_err_mps',
      'vz_err_mps',
      'b_err_m']
```

1.4 Error Statistic [1 point]

We will convert this multi-satellite time series into a single number to make comparison easier. For each variable (e.g., the error in the satellite x ECEF position), we want to take the mean of the absolute value of the error across the entire dataset (i.e., all time and all satellites). Fill in the function for `calculate_error_statistic`.

Hint: In the function, we will pass in the variable across the entire dataset. For example, this could be the satellite x ECEF position error.

Hint: The function should be short (roughly 1 line of code).

ANSWER

[Complete the function below]

1.5 Ablation Study [8 points]

An ablation study is a process of systematically removing parts of a model or algorithm to measure how those removals affect its performance. By seeing how much performance drops when a specific part is removed ("ablated"), you can determine its importance to the overall system.

Ablate *one variable* at a time. In the configuration, only *one* variable should be false, and the rest should be marked true. Record the error for each variable. This table should include the errors for all seven variables $(x, y, z, v_x, v_y, v_z, B)$. The table should span the baseline algorithm above and each ablation. Please include headers for both rows and columns in the table.

Hint: Make sure to include: (1) nothing ablated, (2) only eccentricity ablated, (3) only clock bias and drift ablated, (4) only relativistic clock effects ablated, (5) only secular corrections ablated, (6) only harmonic corrections ablated, (7) only iterative corrections ablated, and (8) only WLS timing corrections ablated.

Below we provide some boilerplate code to run multiple configurations. We create a cache dictionary to avoid recomputing the same data. Image that you start with the config `TFFT` and then you want to run `TFFT` and `TTTT`. The cache will store the data from the first run and then only compute the new data. This can be very useful in your final projects to avoid recomputing the same data over and over. If you make changes to the code and need to recompute the results, just run the cell below to clear the cache.

ANSWER

[Complete the list of configurations below and run the code]

1.6 Error Plots [3 points]

Finally, let's visualize some of the errors from our ablation study. Include the plots for the satellite x ECEF position error for the following ablations:

- only eccentricity ablated
- only secular corrections ablated
- only harmonic corrections ablated

ANSWER

[Provide the requested plots]

1.7 Discussion [4 points]

What are your conclusions from the ablation study? Answer the following questions and hypothesize why they were the case. Some will be easier to explain than others.

- Which three variables most impacted the satellite position estimate? Why might this be the case?
- Which three variables most impacted the satellite velocity estimate? Why might this be the case?
- Which three variables most impacted the satellite clock bias estimate? Why might this be the case?

Hint: You may find it helpful to reference the plots in lecture 5 in your explanation. Using a first principles analysis can also help (e.g., looking the effect of eccentricity or the drift rate of certain variables).

ANSWER

[Answer the questions above. Provide quantitative and qualitative arguments. It is not enough to mention the magnitude of the error without explaining the source.]

Problem 2: Generating PRN Code Replicas [8 points]

Goal of this problem: The GPS signal is overlayed with binary code sequences called Gold codes. In this problem, you will implement the algorithm to create the Gold codes used for every GPS satellite's GPS L1 C/A signal.

First, read these excerpts from Misra and Enge. If you have a physical textbook, these are on pages 62 and 63 at the end of Chapter 2.

Misra and Enge excerpt on binary codes and bit shift registers.

Misra and Enge excerpt on generating the Gold codes for GPS. Note the "phase selector" block will change for different GPS PRNs.

If you look back to problem 2 on homework 1 or problem 3 on homework 2, GPS PRN 10, 28, and 32 are all visible in the sky. So, we will be focusing on these three satellites for this and the next problem to add a bit of realism.

2.1 PRN Sequence [4 points]

Create the entire 1023-chip C/A code for PRN 10. To get the phase selector indices for a particular PRN, you can refer to the most recent GPS Interface Specification (IS-GPS-200 on https://www.navcen.uscg.gov/gps-technical-references) and take a look at the table of Code Phase Assignments **for the C/A signal**.

Report and plot the code's first 16 and last 16 chips in 2 separate plots. Use a stepwise or "staircase" plotting style (e.g., `plt.step` in Python's `matplotlib`), so the signal looks like a binary, switching signal between 0 and 1. Also, ensure that your 2 plots are labeled or titled (to distinguish them), ensure your axes are labeled, and include grid lines.

Hints:

a. The GPS Interface Specification also has the first 10 chips expressed in octal notation (see this conversion table from IBM) so you can check your implementation! For PRN 15, the first ten chips expressed in octal notation are `1131` (the leading digit (1) just represents a "1" for the very first chip, and the last 3 digits correspond to the remaining 9 chips).

b. A simple way to model a shift register in code is with a 10-bit array, e.g., `G = np.concatenate(([new_bit], G[0:9]))` (using 0-indexing).

c. For the sake of this problem and the next problem, you may find it helpful to have a single function whereby you pass the phase selectors for the PRN and the number of chips to generate as input. The function should output the code replica for that satellite up to the number of selected chips.

d. If you want to avoid using XOR operations, you can do the following transformations: (binary $0 \rightarrow +1$) and (binary $1 \rightarrow -1$). Then an XOR operation is equivalent to simple multiplication!

$$
\begin{aligned}
0 \oplus 0 = 0 \quad &\rightarrow \quad +1 \cdot +1 = +1, \\
0 \oplus 1 = 1 \quad &\rightarrow \quad +1 \cdot -1 = -1, \\
1 \oplus 0 = 1 \quad &\rightarrow \quad -1 \cdot +1 = -1, \\
1 \oplus 1 = 0 \quad &\rightarrow \quad -1 \cdot -1 = +1.
\end{aligned}
$$

Just be careful that you *convert back* correctly to binary values when plotting and reporting your chips. And, do not forget to also transform your initialization, correspondingly. *Note*: It is important that a binary 1 is converted to -1 (*not* +1); otherwise this trick will not work.

ANSWER

[Report and plot the code's first 16 and last 16 chips in 2 separate plots. Include your code.]

2.2 Periodic Sequence [2 points]

Create the C/A code output for PRN 10 **from epochs 1024 to 2046** as a 1023-element vector. Report and plot the first 16 and last 16 chips of this vector. How does this code compare with the one you generated in 2.1?

Hint: Be careful with indexing errors when creating the code if using 0-indexing.

ANSWER

[Report and plot the code's first 16 and last 16 chips in 2 separate plots. Include your code.]

2.3 A Different PRN [1 point]

Repeat 2.1 for PRN 28.

Hint: You can check the first 10 Chips Octal values in the GPS Interface Specification.

[Report and plot the code's first 16 and last 16 chips in 2 separate plots. Include your code.]

2.4 A Third PRN [1 point]

Repeat 2.1 for PRN 32.

Hint: You can check the first 10 Chips Octal values in the GPS Interface Specification.

[Report and plot the code's first 16 and last 16 chips in 2 separate plots. Include your code.]

Problem 3: CDMA & Correlation [16 points + 1 bonus]

Goal of this problem: In the first week of class, we briefly talked about how we correlate the incoming signal from GPS with our local replica. With the powerful structure of the Code-Division Multiple Access (CDMA), we can correlate signals despite the very weak power of the GPS signals. This correlation process allows us to (1) detect if a particular satellite signal is present in our incoming signal and (2) evaluate our pseudorange measurement ($\rho^{(k)}$) with remarkable precision. In this problem, we will explore the power of correlation.

First, read the following excerpts from Misra and Enge. If you have a physical textbook, these are on page 64, at the end of Chapter 2.

Misra and Enge excerpt showing alignment of local replica with incoming signal.

Misra and Enge excerpt on auto and cross-correlation.

For this problem, to perform the auto and cross-correlations, you will need to **convert the {0,1} binary sequences from problem 2 to {+1, -1} binary sequences**. (It is customary to transform $0 \rightarrow +1$ and $1 \rightarrow -1$, but it should not matter for this problem since we are only computing auto and cross-correlations.) **For all plots, ensure that your axes are labeled and include grid lines**.

3.1 Autocorrelation [3 points]

Plot the normalized circular autocorrelation function of the code for PRN 10, i.e., $R^{(10)}(n)$. How many unique values of the autocorrelation function are there? What are these unique values?

Hint: Don't forget to normalize (i.e., account for the fraction out front of the sum).

Side-note: Gold codes are pretty special :)

Hint: The circular cross-correlation is defined as:

$$r_{xy}[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \cdot y\left[(n-k) \pmod{N}\right] \quad \text{for } k = 0, 1, \dots, N-1$$

ANSWER

[Compute and plot the cross-correlation, provide its unique values and peak (if present), and explain whether the results are expected.]

3.2 Autocorrelation With Random Noise [2 points]

Create a truly random, length-1023 {+1, -1} binary sequence, where each element is +1 with probability 0.5 and -1 with probability 0.5. Plot its normalized autocorrelation function. How many *unique* values of the autocorrelation function are there?

Hint: Notice in the Misra and Enge excerpt that the example PRN signal is periodic (continuously repeats itself for all time). Assume this is also true for this truly random, length-1023 binary code sequence.

Hint: The number of unique values will change depending on the random code you generate, but it should be fairly different than 1.

Hint: For reproducibility, you can use a `seed` to initialize the randomness the same every time you restart the code.

ANSWER

[Compute and plot the cross-correlation, provide its unique values and peak (if present), and explain whether the results are expected.]

3.3 Delayed Autocorrelation [2 points]

Let's see how we can find the chip offset between the received PRN and the replica PRN. Create a 1023-chip PRN sequence by **circularly** delaying PRN 10 by 300 chips. This delayed signal mimics our received PRN. Plot the normalized cyclic cross-correlation function of this delayed version with PRN 10 (our replica). Is the peak of the correlation where you had expected it to be? Explain why or why not.

Hint: You can exploit the periodic nature of the code sequence and make it cyclic by using a longer code sequence, as in Problem 2.2.

ANSWER

[Compute and plot the cross-correlation, provide its unique values and peak (if present), and explain whether the results are expected.]

3.4 Correlation Between Two PRNs [2 points]

Let's see if we can reject the signals from another satellite. Plot the cyclic cross-correlation between PRN 10 and PRN 28. How does this compare to 3? How many unique values of the cross-correlation function are there, and what are these values?

ANSWER

[Compute and plot the cross-correlation, provide its unique values and peak (if present), and explain whether the results are expected.]

3.5 Composite Signal [2 points]

Now, we have all the pieces to discover the power of CDMA. Let's assess if we can pick out the signal of interest (from PRN 10) from a cluttered signal environment. Create the following three 1023-length PRNs:

1. x_1 as PRN 10 delayed by 200 chips.
2. x_2 as PRN 28 delayed by 150 chips.
3. x_3 as PRN 32 delayed by 300 chips.

Sum these 3 signals together via element-wise addition: $x(n) = x_1(n) + x_2(n) + x_3(n)$. This is our composite signal, mimicking reception from all three satellites that are visible in the sky. Correlate the composite signal x with a replica of PRN 10. Is the peak of the correlation where you had expected it to be? Explain why or why not.

From the lectures, you may recall that the GPS signal is quite weak. Let's see if CDMA can pick out the GPS PRN code even in the presence of radio frequency noise.

ANSWER

[Compute and plot the cross-correlation, provide its peak (if present), and explain whether the results are expected.]

3.6 Adding Noise [2 points]

In order to simulate the effect of radio frequency noise on signal tracking of a C/A code, create a 1023-element vector with noise η, each element of which is independent and identically distributed as a zero-mean Gaussian with a standard deviation of 4. Add this vector of noises to the composite signal from 5. Plot this noisy composite signal and the code for PRN 10 in the same plot to get an idea of the relative amplitude difference (include a legend to distinguish the two lines). Plot for the full 1023-element sequence.

[Compute and plot the code for PRN 10, the composite signal, and the noise.]

3.7 Removing Noise [3 points]

Now, correlate the noisy composite signal from 6 with a replica of PRN 10 and plot the resulting correlation. Is the peak where you would expect it to be? Are you surprised by the ability of PRN 10 to survive the interference from noise and the other PRNs? Explain why or why not.

[Compute and plot the cross-correlation, provide its peak (if present), and explain whether the results are expected.]

Bonus 1 [1 point]

To further demonstrate the power of cross-correlation, let's assume that we are now dealing with a noise standard deviation of 20. Since the PRN codes are periodic, we can treat each period separately. For three different noise realizations, repeat 3.7 and plot the three cross-correlations.

- Are you able to differentiate the peak from the noise? Is the peak where we expect it to be?

Since the noise is typically not correlated over time (this might not always be the case), we can average the cross-correlations obtained during consecutive PRN code periods:

$$R_N[k] = \frac{1}{N} \sum_{i=0}^{N-1} r_{xy}^{(i)}[k]$$

where N is the number of periods we are considering and $r_{xy}^{(i)}$ is the cross-correlation for the i-th period.

- Plot the cross-correlations for an increasing number of considered periods.
- How many sequences do we need to average before the peak is where we expect it to be? How much time would a user have to wait?
- At what rate does the standard deviation of the noise floor decrease as we increase the number of averaged sequences N?

Self-care

We are roughly halfway through the Autumn quarter. And you've made it this far, so congrats! For many of us, we may be experiencing our academic workload picking up in intensity at this point of the quarter. We may correspondingly be tempted to de-prioritize self-care; however, it is when we're at our busiest that it is *especially* important to make time to take care of ourselves.

What is 1 thing, big or small, that you have recently done to take care of yourself? Describe your thoughts in a few sentences. Alternatively, feel free to be creative in your form of expression, including but not limited to, a photo, video, or sketch.

Your Self-Care:

Solution

Answers will vary. Just be yourself!